

Pi-Ener-lite

User Manual

V1.0 -- by XiaoJ

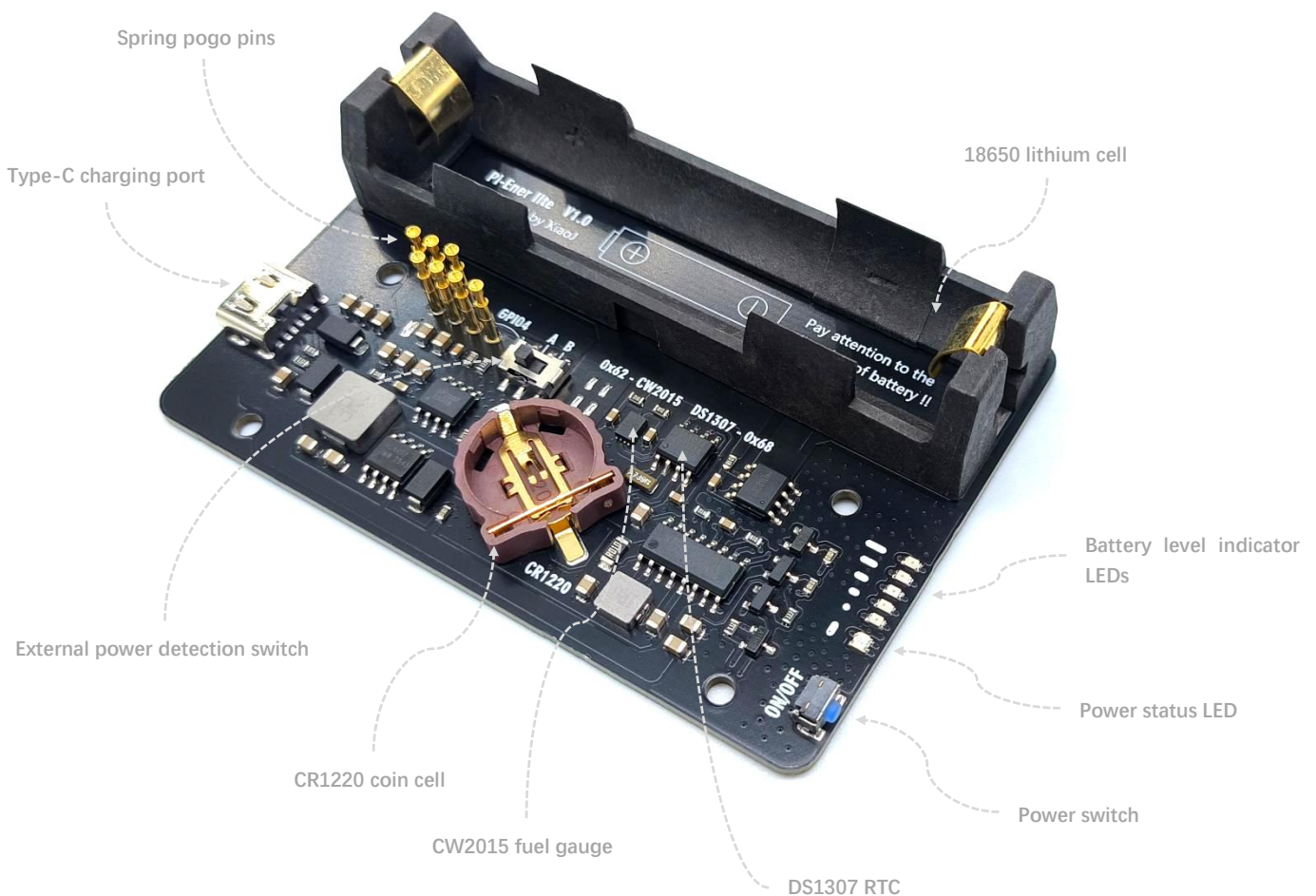


1. Overview

Pi-Ener-lite is a dedicated UPS power supply designed for the Raspberry Pi Zero series motherboards. It connects to the Zero board via 7 spring pogo pins, which handle both power delivery and data reading.

This UPS is powered by a standard flat-top 18650 lithium-ion cell. It seamlessly switches to battery power upon external power loss and supports simultaneous charging and discharging. Built-in protections include overcharge, overdischarge, overcurrent, and short-circuit protection, plus external power insertion detection.

In addition, Pi-Ener-lite integrates the fuel gauge chip CW2015 and the RTC real-time clock chip DS1307. It can measure battery voltage and remaining capacity, and provide RTC timing function for Zero.



Electrical Specifications:

Charging current: **1A@5V**

Discharging output: **2.5A@5.1V**

Battery voltage measurement error: $\pm 3\text{mV}$ (varies slightly by cell batch and ambient temperature)

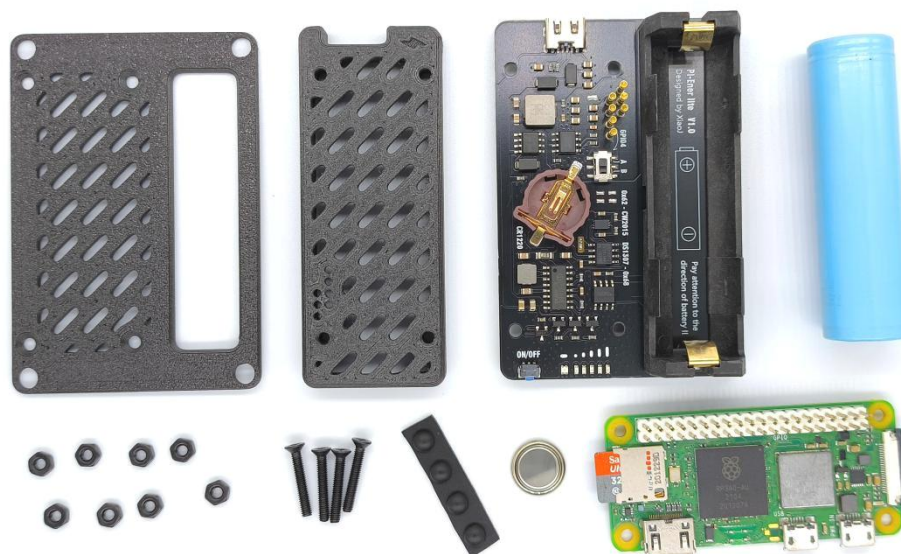
State of Charge (SoC) reading error: $\pm 2\%$

(Note: Use high-quality 18650 cells and power adapters to guarantee stable UPS operation.)

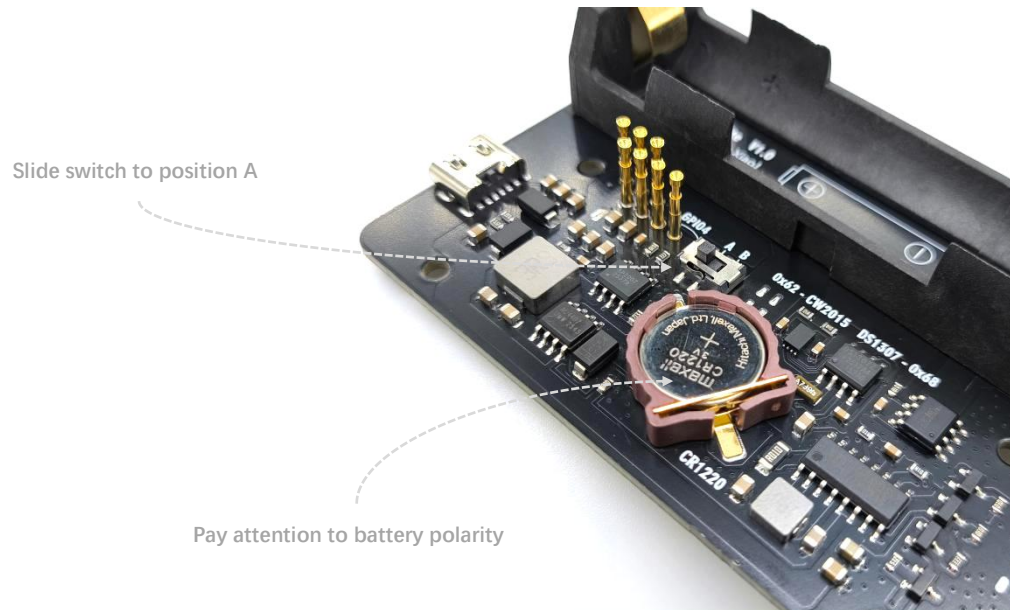
2. Assembly Guide

Please prepare the following materials for assembly

Note: Please prepare your own Zero main board, 18650 battery and CR1220 coin-cell battery



- a. Insert the CR1220 coin cell. Slide the external power detection switch to position "A" to enable the external power insertion detection function



- b. Assemble 3D Printed Enclosure



c. Attach Anti-Slip Silicone Pads



d. Mount Raspberry Pi Zero & 18650 Battery



3. Function Operation Instructions

3.1 Power Switch & LED Indicator Definitions

- **Power Switch Control:** Press and hold the power switch for over 3 seconds to toggle power output ON/OFF.
- **Power Output Indicator LED (Red LED):**
 - Red LED lit = Power output active (ON)
 - Red LED off = Power output disabled (OFF)
- **External Power Indicator (Blue LED):**
 - Blue LED lit = External Type-C power connected
 - Blue LED off = External Type-C power disconnected
- **Battery Level LEDs:** The five battery level LEDs correspond to remaining capacities of 20 %, 40 %, 60 %, 80 % and 100 %, respectively
 - No charging:** Short press the power switch to view current battery level; LEDs auto-turn off after several seconds.
 - Charging state:** Level LEDs flash sequentially to indicate charging progress.
 - Fully charged:** All five battery level LEDs stay solidly lit.

3.2 External Power Insertion Detection

External power insertion can be detected by reading the logic level of GPIO4 on the Zero board. To enable this function, slide the external power detection switch to position "A". This connects GPIO4 to the level-sensing circuit.

A high logic level at GPIO4 means no external power supply is plugged in, while a low logic level indicates external power is present.

If GPIO4 is required for other functions, slide the switch to position "B" to disconnect it from the level-sensing circuit.

3.3 RTC Real-Time Clock Setup

Pi-Ener-lite uses the DS1307 RTC chip for persistent system clock. The CR1220 backup coin cell must be installed correctly; otherwise the RTC will lose time after full power loss. Follow the steps below to configure the RTC on Raspberry Pi OS

- a. Enable the I2C bus via raspi-config, Open the configuration tool in terminal

sudo raspi-config

Navigate to the I2C interface option and enable it.



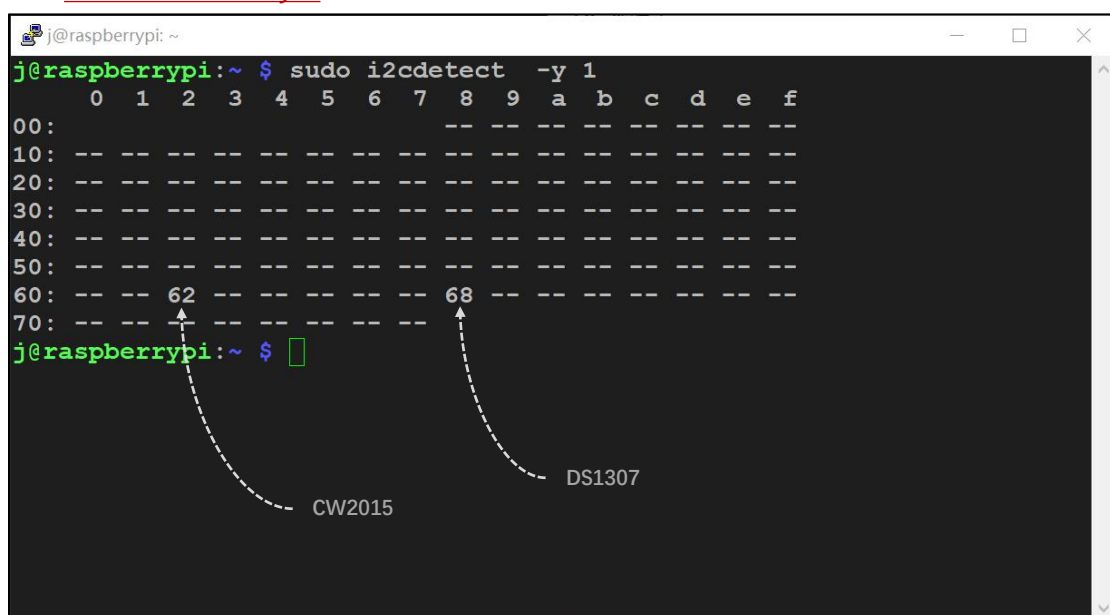
```
j@raspberrypi: ~  
j@raspberrypi:~$ sudo raspi-config
```

- b. Ensure the i2c-tools package is installed on your system. If it is not installed, run the following command to install it

sudo apt install i2c-tools

Run the i2cdetect command to scan I2C devices. The address 0x62 corresponds to CW2015, and 0x68 corresponds to DS1307.

sudo i2cdetect -y 1



```
j@raspberrypi: ~  
j@raspberrypi:~$ sudo i2cdetect -y 1  
 0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f  
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
20:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
40:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
50:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
60:  --  --  62  --  --  --  --  --  68  --  --  --  --  --  --  
70:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
j@raspberrypi:~$
```

CW2015

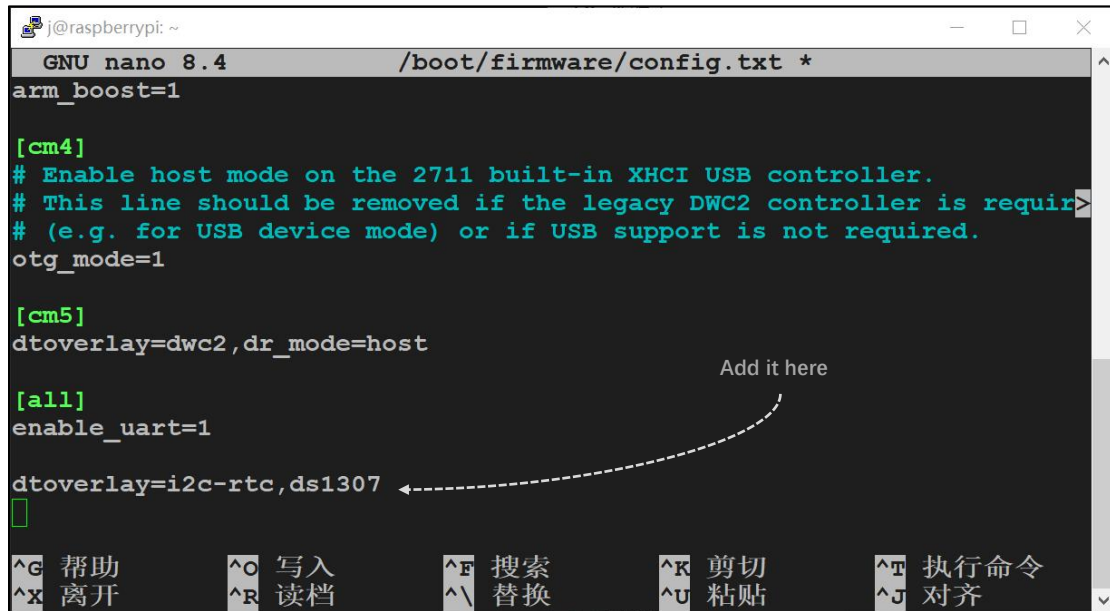
DS1307

c. Use the nano editor to modify the system configuration file config.txt.

sudo nano /boot/firmware/config.txt

Append the following line at the end of config.txt to load the DS1307 device tree overlay.

dtoverlay=i2c-rtc,ds1307



```
GNU nano 8.4 /boot/firmware/config.txt *
arm_boost=1

[cm4]
# Enable host mode on the 2711 built-in XHCI USB controller.
# This line should be removed if the legacy DWC2 controller is required
# (e.g. for USB device mode) or if USB support is not required.
otg_mode=1

[cm5]
dtoverlay=dwc2,dr_mode=host

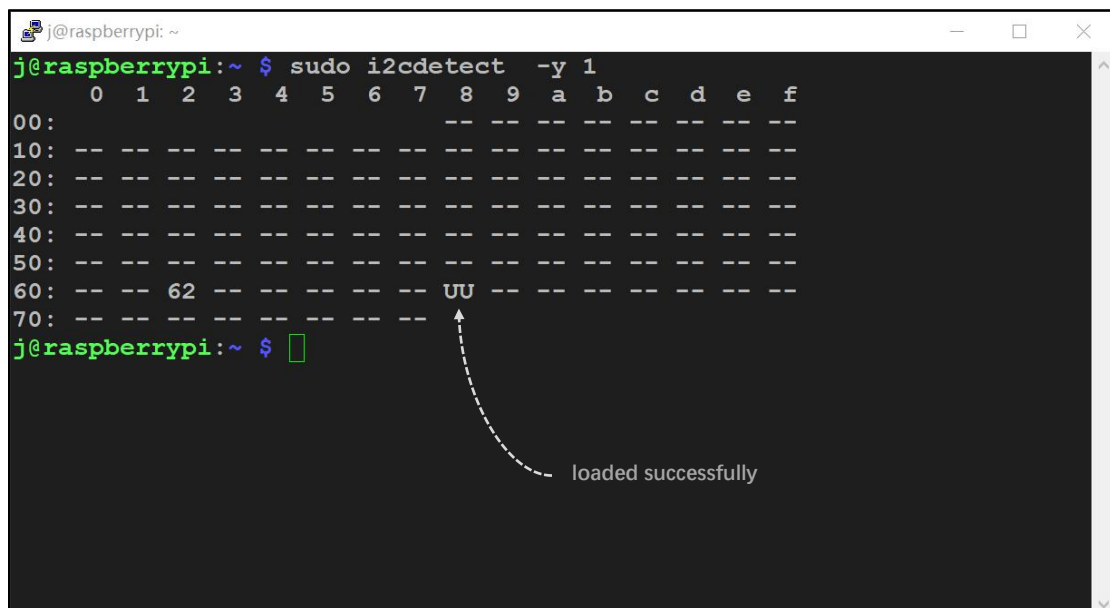
[all]
enable_uart=1

dtoverlay=i2c-rtc,ds1307

```

e. Reboot the system, then run the i2cdetect command to scan I2C devices again. If the address 0x68 shows UU, the DS1307 device has been loaded successfully

sudo i2cdetect -y 1

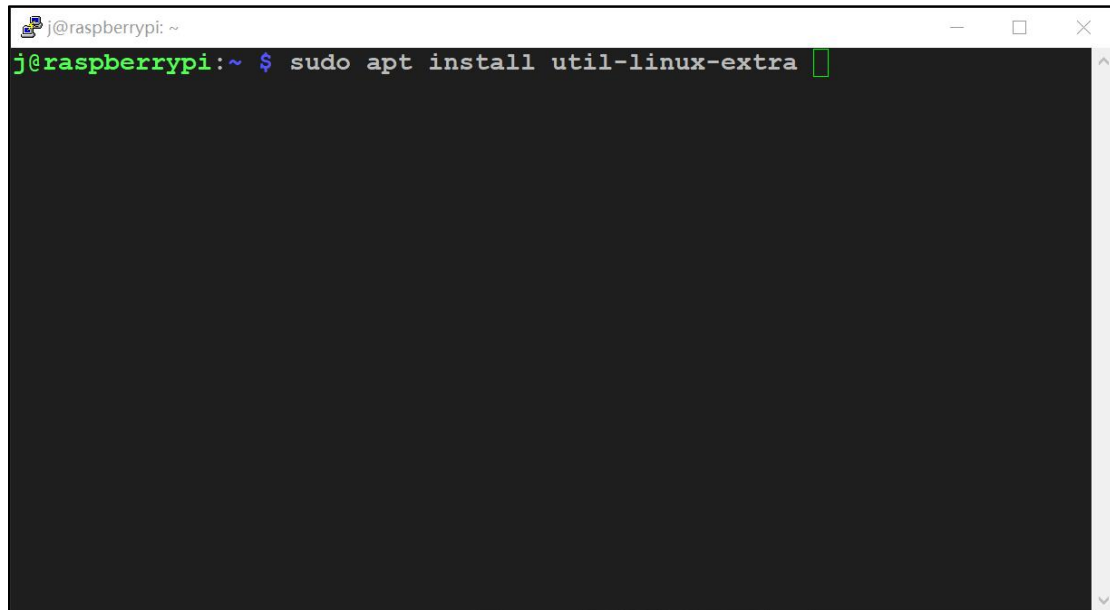


```
j@raspberrypi:~$ sudo i2cdetect -y 1
   0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- 62 -- -- -- -- -- UU -- -- -- -- --
70: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
j@raspberrypi:~$

```


f. Install the util-linux-extra tool package

sudo apt install util-linux-extra

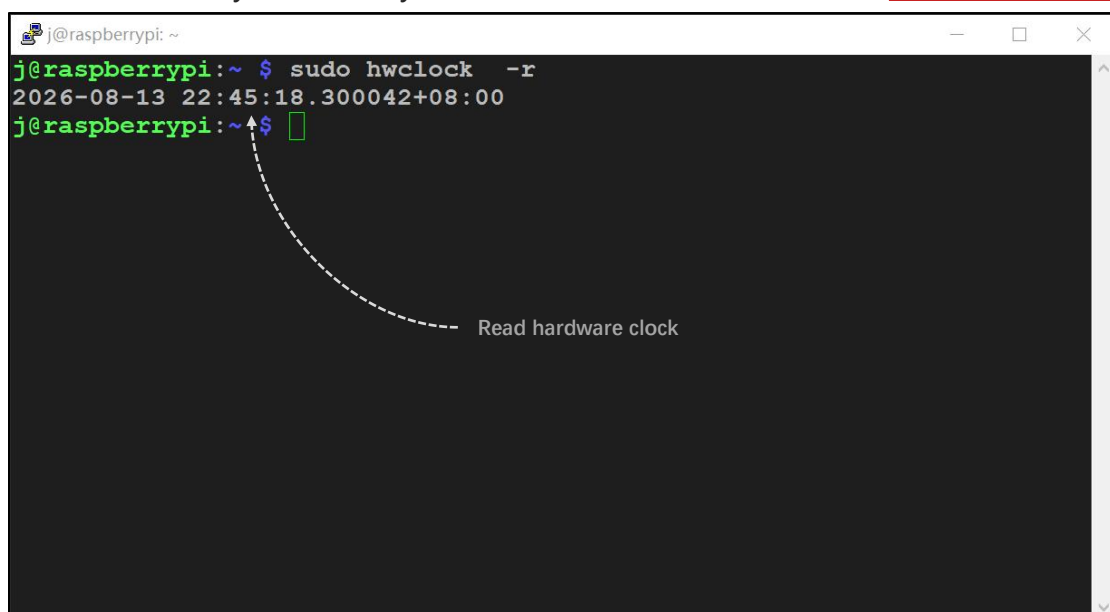


```
j@raspberrypi: ~  
j@raspberrypi:~ $ sudo apt install util-linux-extra
```

g. Use the hwclock command to configure the RTC timing chip. For more configuration options of the hwclock command, please refer to relevant documentation by yourself.

Command for reading the hardware clock: *sudo hwclock -r*

Command to synchronize system time to the hardware clock: *sudo hwclock -w*



```
j@raspberrypi: ~  
j@raspberrypi:~ $ sudo hwclock -r  
2026-08-13 22:45:18.300042+08:00  
j@raspberrypi:~$
```

Read hardware clock

```
j@raspberrypi: ~  
j@raspberrypi:~$ sudo hwclock -w
```

Synchronize hardware clock

3.4 Battery Fuel Gauge Function

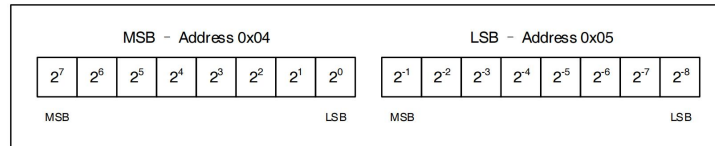
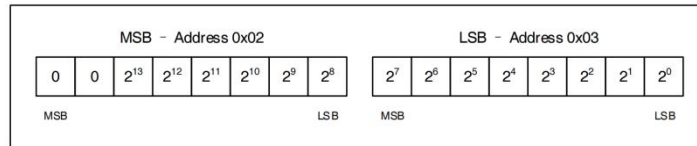
Pi-Ener-lite uses CW2015 for battery voltage and capacity measurement. The Zero communicates with the chip via the I2C bus to obtain battery voltage and remaining-capacity readings.

The device address of CW2015 is 0x62.

The VCELL register holds the 14- bit ADC measurement of battery voltage, located at address 02h- 03h. The ADC resolution is 305 μ V.

The SOC register provides the 16- bit reading of battery capacity percentage, located at address 04h- 05h. The high 8- bit of SOC represents 1% of battery capacity per unit, while the low 8- bit represents 1/256 %, delivering the fractional part of the battery- capacity percentage.

The current battery voltage and remaining capacity can be obtained by reading and calculating data from these two registers. Refer to the provided Python script for detailed calculation procedures. For further information about the CW2015, please consult the official chip datasheet.



Run the reference program to read battery voltage, remaining capacity and external power insertion status.

```
j@raspberrypi: /home/Pi-Ener-Lite $ sudo python Pi-Ener-Lite.py
```

Initialize the CW2015

```
+++++
```

Voltage: 4.17V
Battery: 100%
Battery FULL
Power Adapter Plug In
+++++
+++++

Voltage: 4.17V
Battery: 100%
Battery FULL
Power Adapter Plug In
+++++
+++++

Voltage: 4.17V
Battery: 100%
Battery FULL
Power Adapter Plug In
+++++
+++++

Voltage: 4.17V
Battery: 100%
Battery FULL
Power Adapter Plug In
+++++

```
[
```

The reference script *Pi-Ener-lite.py*

```
#!/usr/bin/env python
import struct
import smbus
import sys
import time
import RPi.GPIO as GPIO

CW2015_ADDRESS    = 0X62
CW2015_REG_VCELL   = 0X02
CW2015_REG_SOC     = 0X04
CW2015_REG_MODE    = 0X0A

def readVoltage(bus):
    "This function returns as float the voltage from the Raspi UPS Hat via
    the provided SMBus object"
    read = bus.read_word_data(CW2015_ADDRESS, CW2015_REG_VCELL)
    swapped = struct.unpack("<H", struct.pack(">H", read))[0]
    voltage = swapped * 0.305 / 1000
    return voltage

def readCapacity(bus):
    "This function returns as a float the remaining capacity of the battery
    connected to the Raspi UPS Hat via the provided SMBus object"
    read = bus.read_word_data(CW2015_ADDRESS, CW2015_REG_SOC)
    swapped = struct.unpack("<H", struct.pack(">H", read))[0]
    capacity = swapped / 256
    return capacity

def QuickStart(bus):
    "This function wake up the CW2015 and make a quick-start fuel-gauge
    calculations "
    bus.write_word_data(CW2015_ADDRESS, CW2015_REG_MODE, 0x30)
```

```

#-----
GPIO.setmode(GPIO.BCM)
GPIO.setwarnings(False)
GPIO.setup(4,GPIO.IN) # GPIO4 is used to detect whether an external power supply
is inserted

bus = smbus.SMBus(1) # 0 = /dev/i2c-0 (port I2C0), 1 = /dev/i2c-1 (port I2C1)

QuickStart(bus)

print (" ")
print ("Initialize the CW2015 .....")

while True:

    print ("+++++")
    print ("Voltage:%5.2fV" % readVoltage(bus))
    print ("Battery:%5i%" % readCapacity(bus))

    if readCapacity(bus) == 100:
        print ("Battery FULL")
    if readCapacity(bus) < 5:
        print ("Battery LOW")

#GPIO is low when power is plugged in
if (GPIO.input(4) == GPIO.LOW):
    print ("Power Adapter Plug In ")
if (GPIO.input(4) == GPIO.HIGH):
    print ("Power Adapter Unplug")

print ("+++++")
time.sleep(2)

```


4. Additional Information

User manuals, reference scripts, enclosure design files and other resources for this project are updated irregularly. The latest materials shall prevail as published on GitHub.

<https://github.com/linshuqin329/Pi-Ener-lite>

Please contact me should any problems arise during operation

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